

Credential-Conditioned Latent Recovery for Wireless Image Semantic Transmission

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Abstract. Wireless image semantic transmission sends continuous latent representations over open wireless channels, where unintended receivers may still recover recognizable image content from the transmitted latent. Existing methods mainly focus on reconstruction quality for legitimate receivers, while controlling who can correctly recover the semantic latent has received less attention. This paper proposes Keyed Latent Recovery (KLR), a credential-conditioned latent access-control mechanism for wireless image semantic transmission. KLR applies credential-derived hard-sign modulation, channel permutation, and spatial token permutation to the transmitted latent. A receiver with the correct credential can generate the matched inverse operations before decoding, whereas an incorrect or missing credential leads to mismatched latent recovery under the shared decoder. KLR does not increase the transmitted latent dimensionality and introduces only lightweight additional computation. Experiments on Kodak and DIV2K under AWGN, together with a Rayleigh-fading sanity check on Kodak, show that KLR preserves legitimate reconstruction quality while substantially reducing unauthorized recoverability. At SNR = 10 dB, Bob achieves an SSIM of 0.8924, while Wrong Key and Eve obtain 0.0520 and 0.0736, respectively, with only a 0.017 dB PSNR decrease relative to the unprotected carrier. Additional wrong-credential statistics, component analyses, and learned-Eve stress tests further examine the behavior of unauthorized recovery. KLR is intended as a credential-conditioned latent recovery mechanism rather than a cryptographic secrecy scheme.

Keywords: Credential-conditioned semantic communication · Keyed Latent Recovery · semantic latent access control · authorized reconstruction · image semantic transmission

1 Introduction

Semantic communication has been regarded as an important candidate technology for future 6G intelligent networks because it aims to transmit task-relevant

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or perception-relevant information rather than only recovering bit-level signals. In wireless image semantic transmission, an image is usually encoded into a continuous semantic latent representation and then reconstructed by a neural decoder at the receiver. However, open wireless channels also make such latent representations observable to unauthorized receivers. Once an eavesdropper or an incorrect-credential receiver can reconstruct recognizable image structures, scene content, or task-related semantic information from the transmitted latent representation, semantic leakage may occur even without recovering a conventional plaintext bitstream. Therefore, this paper focuses on the following problem: in wireless image semantic transmission, how can a receiver with the correct credential recover the image, while receivers with an incorrect credential or without a credential fail to recover effective semantic content?

Existing studies have made important progress in semantic communication and wireless image transmission. DeepSC demonstrates end-to-end semantic coding for text transmission [18], while DeepJSCC and DeepJSCC-f show that neural joint source-channel coding can improve image transmission robustness under noisy channels [1, 8]. WITT further introduces Transformer-based visual modeling into wireless image semantic transmission and improves channel-adaptive reconstruction quality [21]. Meanwhile, recent secure semantic communication studies have pointed out semantic leakage risks from the perspectives of physical-layer security, semantic encryption, adversarial training, and model inversion attacks [9, 11, 3, 13]. Nevertheless, most image semantic transmission methods still mainly optimize the reconstruction quality of the legitimate receiver, while the controlled recoverability of semantic latent representations under different credential conditions remains insufficiently explored. This issue is important because end-to-end image semantic transmission operates on continuous latent representations rather than conventional plaintext bitstreams. Standard digital encryption can protect a discretized bitstream, but inserting a discrete cipher into the continuous latent path introduces a non-differentiable protection interface and is not naturally aligned with the graceful reconstruction behavior of analog-style joint source-channel coding under channel degradation. Our objective is therefore not to replace conventional cryptography, but to study credential-conditioned access control directly on the continuous semantic latent.

To address this issue, this paper presents Keyed Latent Recovery (KLR), a credential-conditioned latent access-control mechanism for wireless image semantic transmission. KLR adopts a WITT-style image semantic transmission structure as an accessible carrier and introduces credential-derived control at the semantic latent layer. A textual credential is mapped to deterministic control parameters, which generate a hard-sign mask, channel permutation, and spatial token permutation to protect the image latent representation. Bob uses the correct credential to generate matched inverse recovery parameters and obtains a decoder-interpretable latent representation. In contrast, Wrong Key uses mismatched recovery parameters, and Eve (No Key) directly decodes the protected latent representation without credential-based recovery. The three receiver branches share the same semantic decoder, so their reconstruction differences

mainly come from whether the latent representation is correctly recovered rather than from different decoder capacities. The focus of this work is not to redesign or improve the WITT backbone itself, but to investigate whether semantic latent recoverability can be controlled by credential-derived keys on top of an image semantic transmission carrier.

Experimental results show that KLR introduces negligible degradation to the legitimate reconstruction quality of Bob while substantially reducing the semantic recoverability of incorrect-credential and no-credential receivers. Under the tested SNR settings, Bob maintains high PSNR and SSIM values close to the Unprotected reference carrier, whereas Wrong Key and Eve (No Key) remain at low structural similarity levels. Further evaluations with multiple wrong credentials, component ablation, and different carrier initialization sources indicate that the recovery failure of unauthorized branches is not caused by a single accidental wrong key or a specific pretrained carrier weight. These results suggest that semantic latent representations in wireless image transmission can be made credential-dependent: the same protected latent representation remains recoverable for an authorized receiver but becomes difficult to interpret for unauthorized receivers. This provides a practical direction for combining semantic communication with credential-based access control at the latent representation level.

2 Related Work

2.1 Image Semantic Transmission

Principle-oriented and survey studies further summarize the architecture, challenges, security, and privacy issues of semantic communication networks [16, 7]. Deep joint source-channel coding is a representative direction in wireless image semantic transmission. DeepJSCC directly maps image pixels to channel input symbols and jointly optimizes source coding, channel transmission, and image reconstruction [1]. Follow-up studies improve this paradigm through feedback transmission [8], attention-based JSCC [19], SNR-adaptive coding [5], and nonlinear-transform source-channel coding [4]. Transformer-based methods further enhance visual representation modeling. WITT introduces Swin Transformer into wireless image semantic transmission [21], and SwinJSCC further investigates Swin-Transformer-based JSCC [20]. These studies mainly focus on improving legitimate reconstruction quality under noisy channels. In contrast, this paper studies how the recoverability of transmitted semantic latent representations can be controlled by credentials.

2.2 Secure Semantic Communication

Security and privacy risks in semantic communication have attracted increasing attention. Existing studies indicate that semantic leakage cannot be fully assessed by bit error rates or bitstream-level secrecy alone, because unauthorized receivers may still recover task-related semantic information [9, 13]. Prior works have explored physical-layer semantic encryption [15], adversarial-training-based privacy

Table 1. Qualitative comparison with representative secure or credential-aware semantic communication approaches. “NR” denotes not explicitly reported under the corresponding formulation.

Method	Control locus	Tx-dim. increase	Credential / control form	Dedicated training	Channel evaluation
ESCS [11]	Semantic encryption	NR	Shared secret key	Yes	AWGN
KGSC-SAT [17]	Decoder gating / feature steganography	NR	Key vector / steganography	Yes	AWGN
SemanAegis [23]	Input / codec backdoor	NR	Input-embedded credential	Yes	AWGN + Rayleigh
SemGuard [22]	Input / codec backdoor	NR	Input-embedded credential	Yes	AWGN + Rayleigh
KLR (ours)	Transmitted semantic latent	No	Locally derived; none transmitted	Yes	AWGN + Rayleigh

preservation [11], coding-enhanced jamming [2], and key-gated secure semantic transmission [17]. Different from methods that primarily rely on an explicitly trained attacker model, a privacy loss, or a channel-quality advantage for Bob, KLR applies credential-conditioned recovery control directly to the transmitted semantic latent representation. The protected latent remains a continuous semantic transmission representation, but its decoder interpretability depends on whether the receiver can generate matched recovery parameters.

2.3 Credential Access Control and Representation Protection

Credential-aware access control is closely related to secure semantic transmission. SemanAegis studies credential-aware semantic communication against knowledge leakage [23], while SemGuard investigates trusted access mechanisms against illegal invocation [22]. Meanwhile, model inversion on semantic communication [3], inversion of deep visual representations [12], and classical model inversion attacks [6] show that intermediate representations may reveal sensitive visual content. Semantic steganography and cross-modal privacy protection also study concealment, recovery, and privacy risks from different perspectives [24, 10, 14]. Unlike these studies, KLR focuses on credential-conditioned recovery of decoder-interpretable latent representations in wireless image semantic transmission. Table 1 summarizes the main architectural differences from representative security and access-control approaches.

The comparison is qualitative because these methods address different threat models and protection objectives, making direct numerical security comparison inappropriate without reimplementing under a unified protocol. KLR differs mainly in applying credential-conditioned reversible control directly to the transmitted latent while keeping its dimensionality unchanged and transmitting no credential-derived mask or permutation indices. Its current channel evaluation covers AWGN and a Rayleigh fading sanity check.

3 System Model and Method

3.1 System Model and Threat Assumption

Let the input image be $x \in \mathbb{R}^{3 \times H \times W}$. A WITT-style semantic encoder E_θ maps the image into a semantic latent representation:

$$z = E_\theta(x), \quad z \in \mathbb{R}^{C \times h \times w}. \quad (1)$$

In ordinary wireless image semantic transmission, the latent representation is distorted by the wireless channel and then reconstructed by the decoder D_ϕ as

$$\hat{x} = D_\phi(H(z)), \quad (2)$$

where $H(\cdot)$ denotes the wireless channel. We argue that modeling only ordinary legitimate transmission is insufficient for characterizing access control in secure semantic image transmission. Although a semantic latent representation is not a conventional plaintext bitstream, it still carries image structures, color distributions, and scene semantics. If an incorrect-credential receiver or an eavesdropper can directly decode such a latent representation, semantic information may still be leaked.

The considered system contains four types of participants. Alice is the sender, who holds the source image x and a textual credential c . Bob is the correct-credential receiver, who holds the same credential as Alice. Wrong Key denotes an incorrect-credential receiver, who holds a formally valid but mismatched credential c' . Eve (No Key) denotes a no-credential eavesdropper, who can observe the protected latent representation transmitted over the wireless channel but does not hold a valid credential. The three receiver branches share the same semantic decoder, so the output differences mainly come from whether the latent representation is correctly recovered rather than from differences in decoder capacity.

This paper considers two main non-adaptive unauthorized recovery scenarios: incorrect-credential recovery and no-credential direct decoding. Wrong Key knows the system structure and shared decoder but uses a mismatched credential c' , whereas Eve (No Key) directly decodes the received protected latent without credential-based recovery. Both therefore use the same decoder capacity as Bob, and their recovery differences arise from credential-matched latent recovery.

We additionally evaluate offline-trained learned Eve attackers in Section 4.5. Here, “non-adaptive” means that the evaluated receiver does not iteratively query the system, refine its strategy from prior attempts, or exploit known/chosen image-protected-latent pairs online. The learned attackers are trained independently without the correct credential and do not perform query-adaptive credential search. Stronger attacks involving large-scale queries, credential-mapping inference, known image-protected-latent pairs, or credential compromise remain outside the current scope.

Table 2 summarizes the receiver capabilities and attack coverage considered in this work.

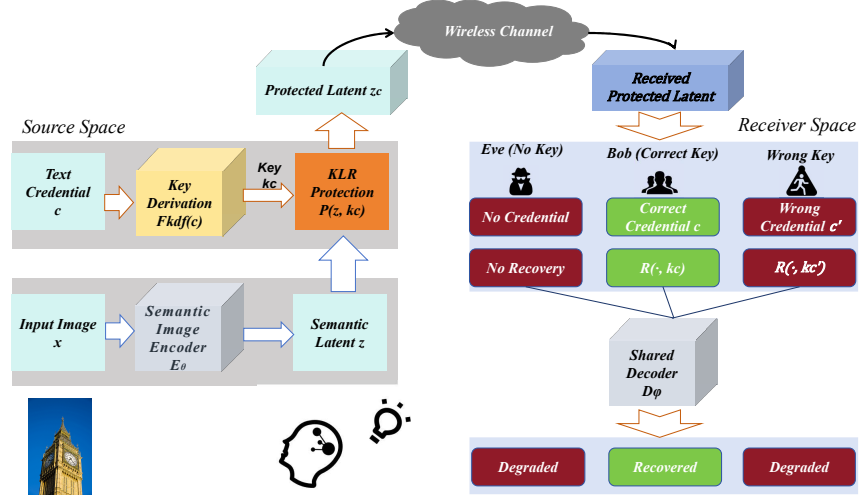


Fig. 1. Overall workflow of the proposed KLR framework. The sender maps the source image into a semantic latent representation and applies credential-derived latent protection. After wireless-channel transmission, Bob uses the correct key to recover a decoder-interpretable latent representation, while Wrong Key and Eve yield degraded reconstructions due to mismatched recovery or no-key direct decoding. The encoder and decoder are implemented with a WITT-style image semantic transmission carrier. The labels k_c and z_c in the figure are simplified visual labels; the corresponding formal symbols are defined in Section 3.2.

3.2 KLR Latent Protection and Recovery Mechanism

Given a textual credential c , KLR uses a deterministic key derivation function to generate a control vector:

$$k_c = F_{\text{kdf}}(c), \quad k_c \in [-1, 1]^{d_c}, \quad (3)$$

where $d_c = 128$ in our experiments. We use counter-based SHA-256 expansion rather than a large text encoder to derive and normalize the control vector to $[-1, 1]$. SHA-256 is used to diffuse local credential changes across the derived values, not as a standalone password-hardening mechanism or cryptographic guarantee. The mapping is deterministic for the same credential, while resistance to credential guessing remains bounded by credential entropy; practical deployments should therefore use high-entropy, randomly generated credentials, and we do not establish a minimum credential length or entropy threshold. Textually similar credentials are treated as distinct and are not assumed to yield similar latent-control parameters or predictable distances between the resulting sign masks or permutations. Section 4.4 empirically tests whether small credential edits lead to partial or gradually improved reconstruction. Under a fixed credential, the derived key deterministically controls the hard-sign mask and channel/spatial permutations under the stated threat model.

Table 2. Threat model and receiver capabilities considered in this work.

Branch / Scenario	Credential	Capability	Role
Bob (Correct Key)	Correct c	Shared decoder; generates matched recovery parameters.	Legitimate quality reference.
Wrong Key	Mismatched c'	Shared decoder; applies mismatched credential-derived recovery.	Non-adaptive unauthorized receiver.
Eve (No Key)	None	Observes the protected latent and directly decodes it without recovery.	Passive eavesdropper.
Learned Eve	None	Offline latent inverter; KLR modules are frozen; no correct credential.	Learned-attack stress test.
Stronger adaptive attacks	Unknown / leaked	May exploit known image-protected-latent pairs, queries, credential guessing, or credential compromise.	Outside current scope.

Based on k_c , KLR generates three types of latent control parameters: a hard-sign mask s_c , a channel permutation π_c^{ch} , and a spatial token permutation π_c^{sp} . Let $\text{Expand}(k_c, L)$ denote repeating and truncating k_c to length L . The channel-wise hard-sign mask is defined as

$$s_c = \text{sgn}_{\pm}(\text{Expand}(k_c, C)), \quad s_c \in \{-1, +1\}^C, \quad (4)$$

where $\text{sgn}_{\pm}(u) = +1$ if $u \geq 0$ and $\text{sgn}_{\pm}(u) = -1$ otherwise. Given a target length L and a salt parameter α , KLR constructs a deterministic sorting score:

$$q_c^{(L, \alpha)}(i) = e_c^{(L)}(i) + \epsilon \sin(i\alpha + \beta e_c^{(L)}(i)), \quad i = 0, \dots, L-1, \quad (5)$$

where $e_c^{(L)} = \text{Expand}(k_c, L)$. In our implementation, $\epsilon = 10^{-4}$ and $\beta = 78.233$. The permutation index is then obtained by

$$\pi_c^{(L, \alpha)} = \text{argsort}(q_c^{(L, \alpha)}). \quad (6)$$

We set $L = C$ to generate the channel permutation π_c^{ch} , and set $L = hw$ to generate the spatial token permutation π_c^{sp} . No runtime randomness is introduced in this process; therefore, Bob can generate the same recovery parameters as the sender as long as he holds the same credential.

In addition to the hard-sign mask, our implementation retains a lightweight SNR-conditioned amplitude term for slight channel-wise amplitude modulation:

$$a_c = s_c \odot [1 + \eta \tanh(g_{\theta}(k_c, \text{SNR}))], \quad (7)$$

where $g_{\theta}(\cdot)$ is a lightweight MLP and $\eta = 0.0175$ in our implementation, obtained from the fixed modulation-strength setting 0.05×0.35 . This amplitude term is mainly retained for compatibility with the existing checkpoint parameter structure and provides only mild modulation. The main access-control effect comes from the hard-sign mask, channel permutation, and spatial token permutation.

The sender-side protection process is formulated as

$$z_c = \Pi_{\pi_c^{\text{sp}}}^{\text{sp}} \left(\Pi_{\pi_c^{\text{ch}}}^{\text{ch}}(z \odot a_c) \right). \quad (8)$$

After wireless-channel transmission, let $\tilde{z}_c = H(z_c)$ denote the received protected latent.

This design exploits the semantic decoder’s dependence on latent values, channel organization, and spatial token arrangement: sign modulation changes feature values, channel permutation changes learned feature-channel ordering, and spatial permutation changes token layout. Each operation is deterministic and admits a matched inverse under the same credential-derived parameters, so Bob can restore the intended latent organization before the shared decoder without increasing latent dimensionality or transmitting additional side information. With an incorrect or missing credential, the latent remains numerically or structurally misaligned. We do not claim novelty for sign modulation or permutation as standalone operations; KLR contributes their credential-conditioned integration for latent recovery together with a shared-decoder protocol separating correct-credential, incorrect-credential, and no-credential recovery.

Bob uses the same credential to generate a_c , π_c^{ch} , and π_c^{sp} , and recovers the latent representation in the inverse order:

$$z_{\text{bob}} = \left[\left(\Pi_{\pi_c^{\text{ch}}}^{\text{ch}} \right)^{-1} \left(\left(\Pi_{\pi_c^{\text{sp}}}^{\text{sp}} \right)^{-1} (\tilde{z}_c) \right) \right] \oslash a_c, \quad (9)$$

where $\oslash$ denotes element-wise division. Wrong Key uses an incorrect credential c' to generate mismatched $a_{c'}$, $\pi_{c'}^{\text{ch}}$, and $\pi_{c'}^{\text{sp}}$, causing the recovered latent representation to be misaligned in both channel semantics and spatial token organization. Eve (No Key) holds no credential and can only directly decode the protected latent representation. In this way, KLR forms an asymmetric recovery difference among Bob (Correct Key), Wrong Key, and Eve (No Key) under a shared decoder.

3.3 Evaluation Metrics and Method Boundary

For each input image x , the system generates $\hat{x}_{\text{bob}}$, $\hat{x}_{\text{wrong}}$, and $\hat{x}_{\text{eve}}$. We use PSNR and SSIM to measure image reconstruction quality, and define

$$S_B = \text{SSIM}(x, \hat{x}_{\text{bob}}), \quad S_W = \text{SSIM}(x, \hat{x}_{\text{wrong}}), \quad S_E = \text{SSIM}(x, \hat{x}_{\text{eve}}). \quad (10)$$

The Recovery Gap is defined as

$$\Delta_{\text{rec}} = S_B - \max(S_W, S_E). \quad (11)$$

PSNR and SSIM measure pixel-level and structural reconstruction quality, respectively, and serve as the main metrics. Because they do not fully capture perceptual or semantic leakage, LPIPS is additionally reported for learned Eve, while CLIP-based similarity provides a complementary semantic-level view.

Under the evaluated threat model, a larger Recovery Gap indicates a stronger structural recovery advantage for Bob over the strongest unauthorized branch; it measures recovery asymmetry rather than information-theoretic or cryptographic security.

To complement structural reconstruction metrics, we further define a lightweight semantic consistency score based on a frozen CLIP image encoder $\Phi(\cdot)$. For a receiver branch r , the semantic consistency between the source image and the reconstructed image is measured as

$$\mathcal{L}_{\text{sem}}^r = \frac{1}{N} \sum_{i=1}^N \cos(\Phi(x_i), \Phi(\hat{x}_i^r)), \quad (12)$$

where $r \in \{\text{Bob}, \text{Wrong}, \text{Eve}\}$. A larger value indicates stronger model-perceived semantic consistency with the source image. For unauthorized branches, a lower value suggests weaker semantic leakage under this CLIP-based proxy. We further define the semantic consistency gap as

$$\Delta_{\text{sem}} = \mathcal{L}_{\text{sem}}^{\text{Bob}} - \max(\mathcal{L}_{\text{sem}}^{\text{Wrong}}, \mathcal{L}_{\text{sem}}^{\text{Eve}}). \quad (13)$$

This score complements reconstruction metrics and is not a formal privacy guarantee.

KLR is evaluated as a credential-conditioned latent recovery mechanism rather than a cryptographic secrecy scheme. The considered attacks include incorrect-credential recovery, no-credential direct decoding, and learned Eve stress tests in Section 4.5; stronger adaptive, query-based, credential-guessing, or credential-compromise attacks remain outside the current experimental scope.

4 Experiments

4.1 Experimental Settings

We use the complete 5,000 images from COCO val2017 as the adaptation training data for KLR, and use Kodak as the main independent test set. Kodak images are not used for training. To reduce the influence of Kodak’s small size, we also evaluate the trained model on the DIV2K validation set with 100 high-resolution images. Unless otherwise specified, the main evaluation uses the AWGN channel. We additionally conduct a Kodak Rayleigh fading sanity check. The main tables report 20 dB, 10 dB, and 5 dB results, and a dense SNR sweep from 0 dB to 20 dB with a 2 dB interval is reported on Kodak.

The WITT-style carrier is initialized with pre-adapted weights and adapted with KLR for one training round. To analyze dependence on carrier initialization, we also evaluate a Random Init carrier with the same WITT-style encoder-decoder structure, trained from random initialization on 5,000 randomly selected COCO train2017 images and evaluated on Kodak. All receiver branches use the same preprocessing, checkpoint, channel setting, power normalization, and metric implementation. The Unprotected reference carrier shares the same encoder, decoder, and transmitted latent dimensionality with KLR, and differs only in whether key-conditioned latent protection and recovery are applied. For Learned Eve, the attacker is trained on a disjoint subset of COCO train2017 while the KLR model remains frozen.

4.2 Kodak Main Results

We first compare KLR-Bob (Correct Key) with the Unprotected reference carrier. This is the most direct baseline for our objective because both paths use the same image semantic transmission carrier and differ only by the proposed key-conditioned latent protection and recovery mechanism. Table 3 reports this controlled comparison under the unified evaluation protocol.

Table 3. Controlled comparison with the unprotected reference carrier. “Ref” denotes the unprotected reference carrier.

SNR	Ref-PSNR	Bob-PSNR	Δ PSNR	Ref-SSIM	Bob-SSIM	Δ SSIM
20	32.14	32.14	-0.001	0.9099	0.9098	-0.0001
10	30.90	30.89	-0.017	0.8927	0.8924	-0.0003
5	28.56	28.58	+0.027	0.8388	0.8389	+0.0000

As shown in Table 3, under all three channel conditions of 20 dB, 10 dB, and 5 dB, the PSNR and SSIM differences between KLR-Bob (Correct Key) and the Unprotected reference carrier are very small. In particular, at 10 dB, the PSNR of KLR-Bob (Correct Key) decreases by only 0.017 dB and the SSIM decreases by about 0.0003. At 5 dB, the PSNR and SSIM of Bob (Correct Key) are slightly higher than those of the Unprotected reference carrier. These results indicate that introducing key-conditioned latent protection and correct-key recovery does not noticeably degrade the image reconstruction quality of the legitimate Bob branch.

Table 4 further reports the main results of Bob (Correct Key), Wrong Key, and Eve (No Key). Bob (Correct Key) maintains high PSNR and SSIM under all three SNR conditions, whereas the structural similarity of Wrong Key and Eve (No Key) remains at a low level.

Table 4. Kodak main results under the unified evaluation protocol

SNR	B-PSNR	W-PSNR	E-PSNR	B-SSIM	W-SSIM	E-SSIM	Gap
20	32.14	10.32	11.69	0.9098	0.0522	0.0768	0.8330
10	30.89	10.31	11.62	0.8924	0.0520	0.0736	0.8188
5	28.58	10.29	11.39	0.8389	0.0515	0.0677	0.7712

From Table 4, Bob (Correct Key) maintains high reconstruction quality at 20 dB, 10 dB, and 5 dB, whereas Wrong Key and Eve (No Key) remain at low SSIM levels. Fig. 2 further reports a denser SNR sweep on Kodak. As SNR increases from 0 dB to 20 dB, Bob SSIM improves from 0.6456 to 0.9098, while Wrong Key and Eve (No Key) remain around 0.050–0.077 SSIM. The Recovery Gap increases

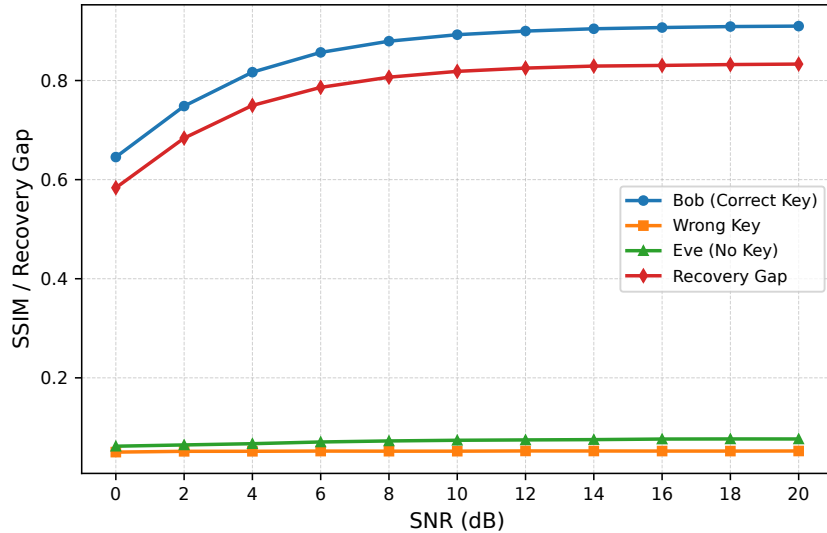


Fig. 2. Dense SNR evaluation on Kodak from 0 dB to 20 dB with a 2 dB interval. Bob (Correct Key) improves with channel quality, while Wrong Key and Eve (No Key) remain at low SSIM levels.

from 0.584 to 0.833, indicating that better channel quality mainly benefits the authorized receiver rather than enabling unauthorized recovery.

To further examine whether the observed separation is limited to the small Kodak dataset, we evaluate KLR on the DIV2K validation set. As shown in Table 5, Bob maintains high reconstruction quality on this larger and more diverse validation set, while Wrong Key and Eve (No Key) stay at very low SSIM levels across all tested SNR values.

Table 5. Generalization evaluation on the DIV2K validation set

SNR	Bob PSNR	Bob SSIM	Wrong SSIM	Eve SSIM	Gap
20	29.78	0.9056	0.0437	0.0532	0.8524
10	28.34	0.8817	0.0434	0.0521	0.8296
5	25.86	0.8131	0.0428	0.0502	0.7629

4.3 Rayleigh Fading Sanity Check

To examine whether the observed authorization gap is limited to the AWGN setting, we further conduct a small-scale sanity check under Rayleigh fading. We use the same Kodak evaluation protocol and the same trained checkpoint without additional retraining. The receiver is assumed to have channel state information and performs simple equalization; CSI estimation error or mismatch is not introduced in this experiment. As shown in Table 6, Bob (Correct Key)

remains very close to the Unprotected reference, while Wrong Key and Eve (No Key) still obtain very low SSIM values. These results indicate that the key-conditioned recovery gap remains observable under the evaluated Rayleigh setting, but do not characterize robustness to imperfect CSI.

Table 6. Rayleigh fading sanity check on Kodak without additional retraining. P and S denote PSNR and SSIM, respectively.

SNR	Base-P	Base-S	Bob-P	Bob-S	Wrong-S	Eve-S	Gap
20	28.37	0.8502	28.45	0.8503	0.0529	0.0737	0.7766
10	27.56	0.8235	27.52	0.8233	0.0528	0.0714	0.7519
5	25.70	0.7501	25.72	0.7501	0.0524	0.0676	0.6825

4.4 Visualization, Ablation, and Wrong-Key Analysis

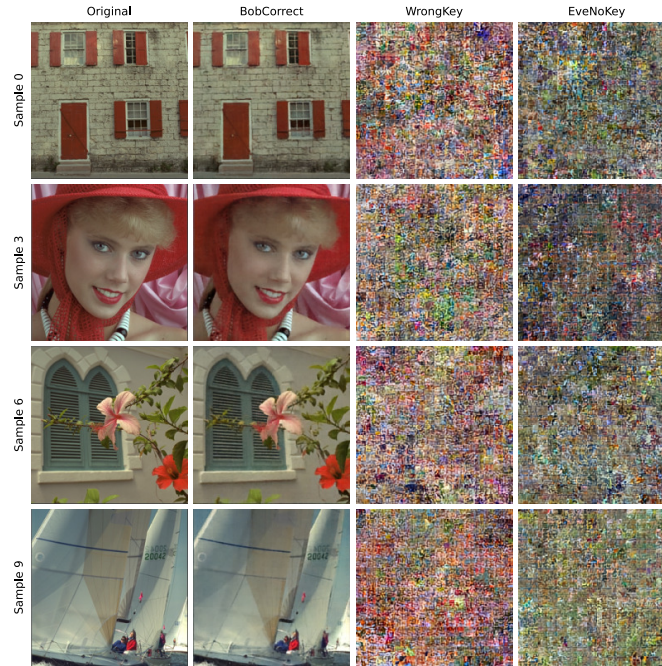


Fig. 3. Qualitative results on Kodak at SNR=10 dB. BobCorrect, WrongKey, and EveNoKey denote Bob (Correct Key), Wrong Key, and Eve (No Key), respectively. Bob preserves recognizable structures, whereas Wrong Key and Eve show obvious degradation.

Fig. 3 shows qualitative results on Kodak at SNR=10 dB. Bob (Correct Key) preserves the main structures and color appearance of the source images, whereas

Wrong Key and Eve (No Key) exhibit strong noise-like and block-like distortions. This visual trend is consistent with the quantitative SSIM and CLIP-Sim results.

To verify whether incorrect-credential suppression depends on a specific wrong key, we generate 100 different wrong credentials at SNR=10 dB. As shown in Table 7, the maximum Wrong Key SSIM is only 0.0775, with a mean of 0.0590 and a standard deviation of 0.0067, far below the Bob SSIM of 0.8925. This indicates that the recovery failure is stable across many wrong credentials rather than being caused by a particular mismatched key.

Table 7. Wrong-credential statistics at SNR=10 dB

Num. Keys	Metric	Min	Mean	Max	Std
100	Wrong Key SSIM	0.0455	0.0590	0.0775	0.0067
100	Δ_{B-w}	0.8150	0.8335	0.8470	0.0067

To further examine whether textual similarity to the correct credential produces partial recovery, we evaluate ten near-miss credentials obtained by small edits to the correct credential. The perturbations include one-character substitution, deletion, insertion, adjacent swap, case change, separator change, additional or trailing spaces, punctuation modification, and a word-level typo. All variants are evaluated on Kodak at SNR=10 dB using the same receiver and decoder setting.

Table 8. Near-miss credential sensitivity on Kodak at SNR=10 dB.

Credential Setting	Count	SSIM	Bob-Branch Gap
Bob (Correct Key)	1	0.8924	Ref.
Near-Miss Credentials	10	0.0564 [0.0497, 0.0612]	0.8361
Random Wrong	1	0.0522	0.8402
Eve (No Key)	1	0.0741	0.8184

As shown in Table 8, the ten near-miss credentials yield a mean SSIM of 0.0564, with all tested values between 0.0497 and 0.0612. This range is close to the random-wrong credential result of 0.0522 and remains far below Bob’s SSIM of 0.8924. Under the evaluated setting, small textual changes therefore do not produce a gradual recovery toward the correct-credential branch. Together with the hash-based derivation, this result suggests that the tested local credential edits do not provide a useful gradual recovery signal. However, this empirical observation should not be interpreted as a formal avalanche guarantee at the permutation level or as resistance to exhaustive or dictionary-based credential guessing.

To further analyze the role of different latent protection components, we conduct a component ablation and internal security-baseline comparison at SNR=10 dB. These variants also serve as internal protection baselines, because they compare the full KLR design with simpler latent protection strategies under the same carrier, dataset, channel setting, and evaluation protocol. Table 9 reports the SSIM results under different component combinations.

Table 9. Component ablation and internal security-baseline comparison at SNR=10 dB.

Variant	B-SSIM	W-SSIM	E-SSIM	Δ_{rec}
Full KLR	0.8922	0.0523	0.0743	0.8179
Sign only	0.8926	0.0360	0.0970	0.7956
Channel only	0.8924	0.0542	0.0684	0.8240
Spatial only	0.8923	0.2432	0.2463	0.6460
Sign + Channel	0.8926	0.0549	0.0893	0.8033
Sign + Spatial	0.8926	0.0525	0.0817	0.8109
Channel + Spatial	0.8924	0.0545	0.0627	0.8297

All variants keep Bob SSIM at about 0.892, indicating that correct-key recovery can largely preserve legitimate reconstruction quality. Spatial only leaves much higher Wrong Key and Eve (No Key) SSIM values, which suggests that disrupting spatial token arrangement alone is insufficient to stably suppress unauthorized recovery. Sign modulation more directly changes the numerical sign distribution, while channel permutation disrupts the feature-channel organization expected by the decoder. It is worth noting that Full KLR does not achieve the largest numerical Recovery Gap among all component combinations. However, Full KLR provides a balanced trade-off between suppressing incorrect-key recovery and no-key direct decoding while preserving Bob reconstruction quality. This suggests that the three components play complementary roles rather than monotonically improving a single metric.

To complement the SSIM-based Recovery Gap, we additionally compute CLIP-Sim on Kodak at SNR=10 dB. Bob (Correct Key) obtains a high CLIP-Sim of 0.9521 ± 0.0219 , while Wrong Key and Eve (No Key) obtain much lower values of 0.5667 ± 0.0481 and 0.5550 ± 0.0541 , respectively. This leads to a semantic consistency gap of 0.3854, indicating that unauthorized branches also lose substantial model-perceived semantic consistency under this CLIP-based proxy.

4.5 Learned Eve Attack Analysis

We further evaluate learned no-credential Eve attackers at SNR = 10 dB. Unlike Direct Eve, which directly decodes the protected latent, learned Eve trains an auxiliary network to map the received protected latent to an estimated decoder-interpretable latent while the KLR encoder, decoder, and protection module remain frozen. The attackers are trained on the same disjoint COCO train2017

Table 10. Learned Eve attack results at SNR = 10 dB under a unified evaluation protocol.

Receiver / Attack	Params	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Bob (Correct Key)	–	30.90	0.8925	0.0977
Direct Eve (No Key)	0M	11.62	0.0737	0.9165
Lightweight Learned Eve	2.14M	14.05	0.4527	0.9020
Strong Learned Eve	257.29M	18.28	0.5066	0.8003

subset without access to the correct credential and are evaluated on Kodak. We compare a lightweight residual-CNN inverter (2.14M parameters) with a substantially larger U-Net-style latent inverter (257.29M parameters).

As shown in Table 10, the higher-capacity U-Net-style attacker achieves better unauthorized reconstruction than the lightweight residual-CNN attacker. The lightweight attacker raises SSIM from 0.0737 for Direct Eve to 0.4527, while the strong attacker further increases SSIM to 0.5066 and PSNR to 18.28 dB. Nevertheless, the strong attacker remains substantially below Bob, whose SSIM is 0.8925 and LPIPS is 0.0977 under the same setting. These results indicate that attacker model capacity and architecture can materially affect unauthorized reconstruction. However, they should not be interpreted as a general security guarantee: under the evaluated training data, attack formulation, and model capacities, the higher-capacity attacker improves recovery but does not approach correct-credential reconstruction quality.

4.6 Carrier Initialization Source Analysis

To examine whether KLR depends on a specific pre-adapted carrier, we further evaluate a Random Init carrier with the same WITT-style encoder-decoder structure but without loading WITT pre-adapted weights. This carrier is trained from random initialization on 5,000 randomly selected COCO train2017 images, validated on COCO val2017, and evaluated on Kodak.

As shown in Table 11, Random Init reduces Bob reconstruction quality compared with Pre-adapted Init, but still obtains usable transmission performance at SNR=10 dB. More importantly, Wrong Key and Eve (No Key) remain far below Bob, with a Recovery Gap of 0.7668. This suggests that KLR is not limited to a specific pre-adapted checkpoint and can still form an authorized-unauthorized recovery gap on a self-trained WITT-style carrier.

Table 11. KLR results under different carrier initialization sources

Carrier Init	SNR	B-PSNR	B-SSIM	W-SSIM	E-SSIM	Gap
Pre-adapted Init	10	30.89	0.8924	0.0520	0.0736	0.8188
Random Init	10	28.45	0.8795	0.1067	0.1127	0.7668

4.7 Computational Complexity and Communication Overhead

KLR keeps the transmitted latent dimensionality unchanged. In our evaluated model, the latent tensor has a shape of $96 \times 16 \times 16$, corresponding to 24,576 latent symbols per image, the same as the unprotected carrier. The credential, hard-sign mask, and permutation indices are not transmitted over the wireless channel. For a fixed credential, masks and permutation indices can be precomputed, and the additional computation mainly consists of element-wise modulation/division and tensor gathering. The lightweight KLR modulation network contains 0.148M parameters, accounting for 0.52% of the 28.35M model, and the single-image latency increases from 17.78 ms to 18.94 ms at SNR=10 dB, corresponding to about 6.53% overhead.

4.8 Discussion and Limitations

Functionally, KLR is a latent-level perceptual access-control mechanism rather than an information-theoretic cipher: the protected latent remains continuous and may retain exploitable structure, as reflected by learned Eve, while the correct credential restores the organization expected by the shared decoder.

Several limitations remain. The current evaluation mainly considers AWGN and a Kodak Rayleigh-fading sanity check; CSI mismatch, broader fading conditions, and time-varying channels require further study. KLR also assumes that Alice and Bob receive the same high-entropy secret through an external credential-management layer; credential generation, distribution, secure storage, revocation/rekeying, and multi-user management are outside the current system design. Low-entropy credentials remain vulnerable to guessing or dictionary attacks. Query-based attacks, attacks with known image-protected-latent pairs, and other semantic transmission architectures also remain future work. Accordingly, the reported results characterize credential-conditioned recoverability under the evaluated threat models, not a formal cryptographic security guarantee.

5 Conclusion

This paper presents KLR, a credential-conditioned latent recovery mechanism for wireless image semantic transmission. By applying credential-derived sign modulation, channel permutation, and spatial token permutation at the semantic latent layer, KLR preserves authorized reconstruction quality while suppressing Wrong Key and Eve under the evaluated threat models. Experiments on Kodak and DIV2K, together with channel, ablation, learned-Eve, initialization, and complexity analyses, demonstrate the feasibility of latent-level credential-conditioned access control without increasing the transmitted latent dimensionality.

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